

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	3+1	9	18

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

We have, $X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}$

Let LSE of θ_1 be $\hat{\theta}_1$ and LSE of θ_2 be $\hat{\theta}_2$

$$\therefore \text{Let } \hat{\beta} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}$$

We know,

$$\hat{\beta} = (X^T X)^{-1} X^T y \text{ where } y = [y_1 \ y_2 \ y_3]^T$$

are the observed values of the response variable.

$$\text{Now, } (X^T X)^{-1} X^T = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{2}{9} \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} \frac{y_1}{3} + \frac{y_3}{3} \\ \frac{y_1}{3} - \frac{y_2}{3} \end{bmatrix}$$

$$\therefore \hat{\theta}_1 = \frac{y_1 + y_3}{3}, \quad \hat{\theta}_2 = \frac{y_1 - y_2}{3}.$$

$$\therefore \text{LSE of } \theta_1 = \frac{y_1 + y_3}{3}, \quad \text{LSE of } \theta_2 = \frac{y_1 - y_2}{3},$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.
- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

(a) We have, $\mathbf{X} = [U_1 \ U_2 \ U_1 + U_2]'$. Since $U_1, U_2 \sim \text{Uniform}(0, 1)$
 $E(U_1) = E(U_2) = \frac{1}{2}$, $E(U_1 + U_2) = E(U_1) + E(U_2) = 1$
 $\text{Var}(U_1) = E(U_1^2) - (E(U_1))^2$ ~~$[U_1, U_2 \text{ are independent}]$~~

Now, $E(U_1^2) = \int_0^1 x^2 dx = \frac{1}{3}$ $\therefore \text{Var}(U_1) = \text{Var}(U_2) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}$

$\therefore \text{Var}(U_1 + U_2) = \text{Var}(U_1) + \text{Var}(U_2) = \frac{1}{6}$ $[\text{Cov}(U_1, U_2) = 0 \text{ as they are independent}]$

Now, $\rho_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sqrt{\text{Var}(X_i)} \sqrt{\text{Var}(X_j)}}$. We have $\text{Cov}(U_1, U_2) = 0$
 $\therefore \rho_{12} = \rho_{21} = 0$

Also, $\text{Cov}(U_1, U_1) = \text{Cov}(U_2, U_2) = \text{Var}(U_1) = \text{Var}(U_2) = \frac{1}{12}$

$\therefore \rho_{11} = 1, \rho_{12} = 0, \rho_{13} = \frac{\text{Cov}(U_1, U_1 + U_2)}{\sqrt{\text{Var}(U_1)} \sqrt{\text{Var}(U_1 + U_2)}} = \frac{\text{Cov}(U_1, U_1)}{\sqrt{\frac{1}{12}} \cdot \sqrt{\frac{1}{6}}} = \frac{1}{\sqrt{2}}$

$\rho_{21} = 0, \rho_{22} = 1, \rho_{23} = \frac{1}{\sqrt{2}}$

$\rho_{31} = \frac{1}{\sqrt{2}}, \rho_{32} = \frac{1}{\sqrt{2}}, \rho_{33} = 1$

$\therefore \rho = \begin{bmatrix} 1 & 0 & 1/\sqrt{2} \\ 0 & 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1 \end{bmatrix}$ is the correlation matrix of $\mathbf{X}$.

(b) ~~we~~ We find the eigenvalues of ρ using $\rightarrow$

$\det(\rho - \lambda I) = 0 \Rightarrow \det \begin{bmatrix} 1-\lambda & 0 & 1/\sqrt{2} \\ 0 & 1-\lambda & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1-\lambda \end{bmatrix}$

$$\Rightarrow (1-\lambda) \left[(1-\lambda)^2 - \frac{1}{2} \right] - \frac{1}{2} (1-\lambda) = 0$$

$$\Rightarrow (1-\lambda) \left[(1-\lambda)^2 - 1 \right] = 0$$

$$\Rightarrow (1-\lambda) (2-\lambda) (-\lambda) = 0$$

$$\Rightarrow \lambda (\lambda-2) (\lambda-1) = 0$$

Here, $\lambda = 2$ is the largest eigenvalue.

$\therefore$ Variance of the first principal component

$$= \sqrt{2}$$

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3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

i. Complete linkage hierarchical procedure.

ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

→ Complete linkage hierarchical procedure
 (a)(i) We have, $D_0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 11 & 5 \\ 1 & 0 & 2 & 3 \\ 11 & 2 & 0 & 3.5 \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{matrix}$ → The minimum entry (distance) is 1 (distance between 1 and 2)

→ Rebuild the matrix after merging 1 & 2 →

$D_1 = \begin{matrix} & \begin{matrix} (1,2) & 3 & 4 \end{matrix} \\ \begin{matrix} (1,2) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 11 & 5 \\ 11 & 0 & 3.5 \\ 5 & 3.5 & 0 \end{bmatrix} \end{matrix}$ [Since, in complete linkage hierarchical procedure, $d_{(1,2),3} = \max(d_{1,3}, d_{2,3})$]

→ The min. entry is 3.5 (distance between 3 and 4), so we merge them as →

$D_2 = \begin{matrix} & \begin{matrix} (1,2) & (3,4) \end{matrix} \\ \begin{matrix} (1,2) \\ (3,4) \end{matrix} & \begin{bmatrix} 0 & 11 \\ 11 & 0 \end{bmatrix} \end{matrix}$

→ Thus final merging height will be 11 → $((1,2), (3,4))$ ← merge

(a)(ii) Avg. linkage hierarchical procedure →

We have, $D_0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 11 & 5 \\ 1 & 0 & 2 & 3 \\ 11 & 2 & 0 & 3.5 \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{matrix}$ → Min. entry is 1 (distance between 1 and 2)

→ Rebuild the matrix after merging 1 & 2 →

$D_1 = \begin{matrix} & \begin{matrix} (1,2) & 3 & 4 \end{matrix} \\ \begin{matrix} (1,2) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 13/2 & 4 \\ 13/2 & 0 & 3.5 \\ 4 & 3.5 & 0 \end{bmatrix} \end{matrix}$ [Since in avg. linkage, $d_{(1,2),3} = \text{avg.}(d_{1,3}, d_{2,3})$]

→ Min entry is 3.5 (distance between 3, 4), so we merge them as $\overrightarrow{(1,2)} \quad (3,4)$

$$D_2 = \begin{matrix} (1,2) \\ (3,4) \end{matrix} \begin{bmatrix} 0 & 5.25 \\ 5.25 & 0 \end{bmatrix}$$

[We took all pairwise distances and took avg, i.e.,

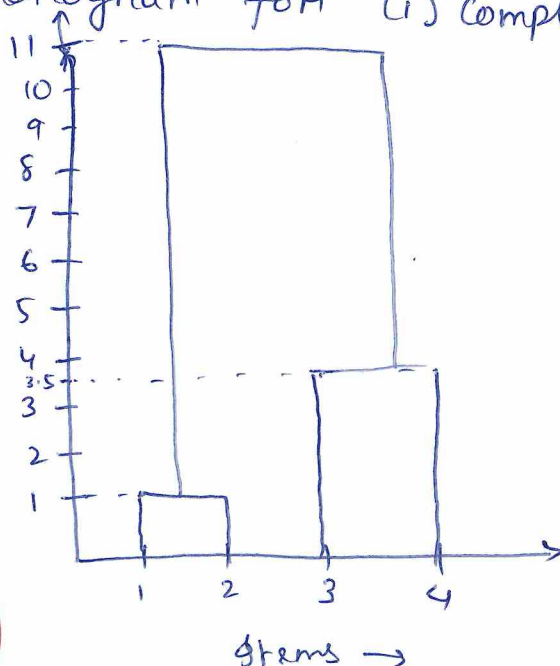
$$\frac{d_{1,3} + d_{1,4} + d_{2,3} + d_{2,4}}{4}]$$

D3

Thus final merging height will be 5.25 → $((1,2), (3,4)) \leftarrow \text{merge}$.

(b) Dendrogram for (i) Complete Linkage →

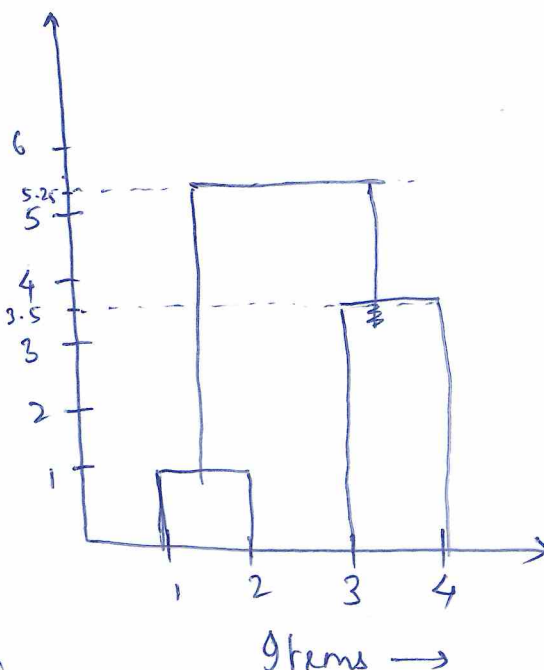
↑
Heights at which items are merged



D3

(ii) Avg. Linkage →

↑
Heights at which items are merged



Observation: We see that the first and second merging are at identical heights but in case of complete linkage, though the two clusters merge (at the last step) with a reasonably larger distance (height of 11) but in case of average linkage the last cluster forms at a much lesser distance (height of 5.25)

ADDITIONAL PAGES FOR ANSWERS

